

Unemployment Insurance, Disability Insurance, and Workers' Compensation

Danny Yagan
UC Berkeley
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Poll (No Right Answer)

A person who is poor because they just lost their job should get a larger transfer than someone who is equally poor and has been that poor for a long time.

- A. True
- B. False
- C. Uncertain

When the government gives transfers to unemployed people, they are less likely to find a job.

- A. True
- B. False
- C. Uncertain

Institutional Features

- Unemployment insurance, workers' compensation, and disability insurance are three social insurance programs in the United States, and they share many common features.
- Unemployment insurance (UI): A federally mandated, state-run program in which payroll taxes are used to pay benefits to unemployed workers laid off by companies.
- Disability insurance (DI): A federal program in which a portion of the Social Security payroll tax is used to pay benefits to workers who have suffered a medical impairment that leaves them permanently unable to work.
- Workers' compensation (WC): State-mandated insurance, which firms generally buy from private insurers, that pays for medical costs and lost wages associated with an on-the-job injury.

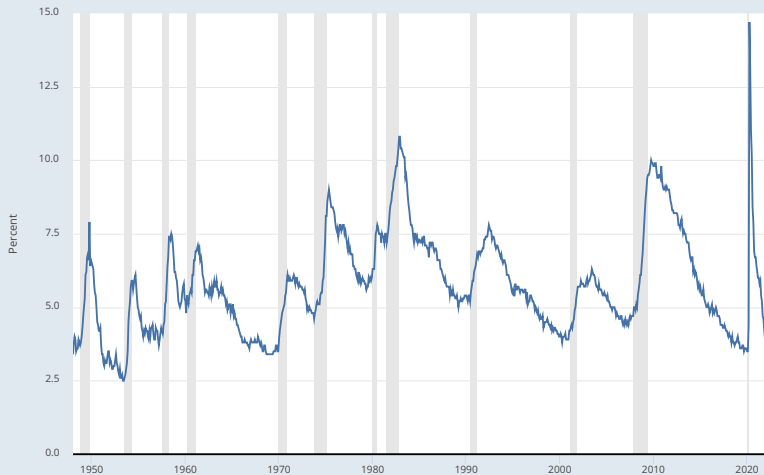
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Comparison of the Features of UI, DI, and WC

Characteristic	UI	DI	WC
Qualifying Event	Job loss, job search	Disability	On-the-job injury
Duration	26-65 weeks	Indefinite	Indefinite (if verified)
Difficulty of verification	Job loss: easy Search: impossible	Somewhat difficult	Very difficult
Average after tax replacement rate	47%	60%	89%
Variation across states	Benefits and other rules	Only disability determination	Benefits and other rules

Unemployment Insurance

- Unemployment insurance is a major social insurance program in the U.S.
- Spending size: \$50bn/year in normal times (up to \$150bn/year during Great Recession)
- Macroeconomic importance in stabilization/stimulus
- Like other social programs, triggered by an event — in this case, involuntary job loss
- Controversial debate about unemployment benefits:
 - Benefit: helps people in a time of need
 - Cost: reduces incentive to search for work while unemployed



Shaded areas indicate U.S. recessions.

Source: U.S. Bureau of Labor Statistics

fred.stlouisfed.org

Institutional Features of Unemployment Insurance

- UI is a federally mandated, state-run program
- Although UI is federally-mandated, each state sets its own parameters on the program.
- This creates a great deal of variation across states — useful as a “laboratory” for empirical work $\Rightarrow$ UI is a heavily studied program
- In 2020 crisis, most state systems unable to cope with volume and new expanded rules $\Rightarrow$ Weakness of decentralized system

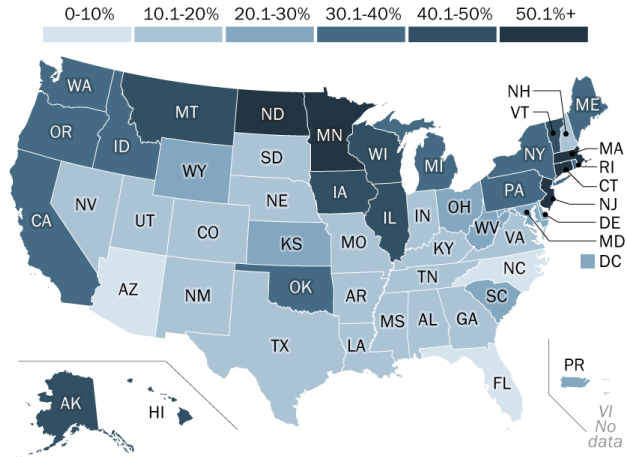
Financing of UI Benefits

- UI is financed through a payroll tax on employers:
 - An employee will not see a deduction for UI on his or her paycheck
 - This payroll tax averages 1-2% of earnings
- UI is partially experience-rated on firms:
 - The tax that finances the UI program rises as firms have more layoffs, but not on a one-for-one basis

Eligibility Requirements and Benefits

- Individuals must have earned a minimum amount over the previous year.
- Unemployment spell must be a result of a layoff, rather than from quitting or getting fired for cause (easy to check)
- Individual must be actively seeking work and willing to accept a job comparable to the one lost (hard to check)
- These eligibility requirements mean that not all of the unemployed actually collect benefits.
- Even among eligible, 50% do not take up the UI benefit (lack of information about eligibility, stigma from collecting a government handout, or transaction costs)
- Take-up typically lower in good times and depends on how hard states make enrollment (e.g. Florida makes it hard)

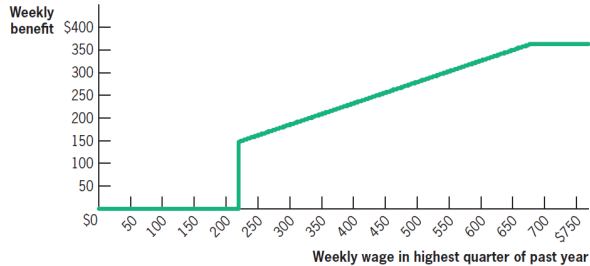
Share of state's unemployed workers receiving unemployment benefits, March 2020



- UI benefits are a function of previous earnings. These benefits vary by state.
- The replacement rate is the amount of previous earnings that is replaced by the UI system: $R = B/W$
- Replacement rates vary from 35% to 55% of earnings
- In 2020 coronavirus crisis, CARES increased weekly benefits by \$600 across the board for 4 months, and expands eligibility to self-employed and lower earners
- Average UI benefit jumps up from \$400 to \$1000/week. Person on \$10/hour wage making \$400/week now makes more on UI. Uniform \$600/week done bc of admin simplicity

14.1

Unemployment Benefit Schedule for Michigan



UI Benefits Duration

- In general, one can collect UI for 6 months.
- In recessions, benefits are automatically extended to 9 months or 12 months
- In deep recessions, benefits can be further extended (23 months in 2008-13)
- Duration of UI benefits typically much higher in European countries
- In 2020 coronavirus crisis, CARES increased duration to 9 months, likely to be extended further as unemployment rate will jump up to 20% and stay above 10% likely for 1-2 years.
- EU countries tend to have more generous and longer benefits

Analysis of Optimal Unemployment Insurance

- Optimal UI trades-off insurance value vs. efficiency costs
- In principle, provide full insurance (perfect consumption smoothing) with 100% replacement rate if there is no moral hazard
- With moral hazard, 100% replacement rate would eliminate incentives to find a job
- $\Rightarrow$ Optimal replacement rate should be less than 100%
- Optimal replacement rate depends negatively on the size of moral hazard and positively on how much people value insurance
- Empirical work examines size of moral hazard and value of UI for consumption smoothing

Expected Utility Model

- Individual's expected utility: $EU = (1 - p)u(c_e) + pu(c_u) = (1 - p)u(w - t) + pu(b)$
- p : probability of being unemployed
- c_e = consumption when employed, c_u = consumption when unemployed
- w = wage when working, t = tax used to finance program, b = UI benefit
- Government needs to balance budget (taxes fund benefits):
$$(1 - p) \cdot t = p \cdot b \quad \Rightarrow \quad t = (p / (1 - p)) \cdot b$$

Optimal UI with No Moral Hazard

- No moral hazard means that p is not affected by UI
- Plugging in govt. budget constraint, rewrite individual's expected utility as:
$$EU = (1 - p)u(w - (p/(1 - p))b) + pu(b)$$
- Government's problem: find b that maximizes EU .
- Optimal benefit b^* will be b such that: $c_u = c_e$
- This is full insurance (as we saw earlier in class)

Optimal UI with Moral Hazard

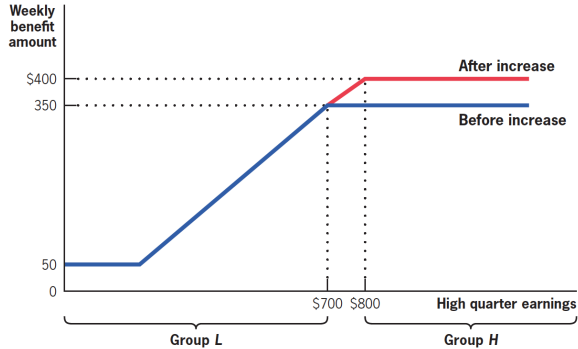
- With moral hazard, p increases with b as more generous benefits deter job search and hence increase unemployment
- Government now chooses b to maximize EU but taking into account that p is a function of b in the budget constraint: $EU = (1 - p)u(w - [p(b)/(1 - p(b))]b) + pu(b)$
- Get new formula: $\frac{u'(c_u) - u'(c_e)}{u'(c_e)} = \frac{1}{1-p} \varepsilon_{p,b}$ with $\varepsilon_{p,b} = \frac{b}{p} \cdot \frac{dp}{db}$
- $\varepsilon_{p,b} > 0$ is the elasticity of unemployment rate with respect to benefits (captures size of moral hazard effects)
- Now $0 < c_u < c_e < w$: partial insurance is optimum. Optimum level increases with curvature of $u(\cdot)$ but decreases with elasticity $\varepsilon_{p,b}$.

Empirical Estimation of Effects of UI

- Moral hazard in UI manifests itself in the duration of the unemployment spell
- Economists ask whether the unemployed find jobs more slowly when benefits are higher
- Key challenge: need to use quasi-experiments to identify these effects
- One common empirical approach (Meyer 1990): difference-in-difference
- Exploit changes in UI laws that affect a “treatment” group and compare to a “control” group

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EVIDENCE: Moral Hazard Effects of Unemployment Insurance

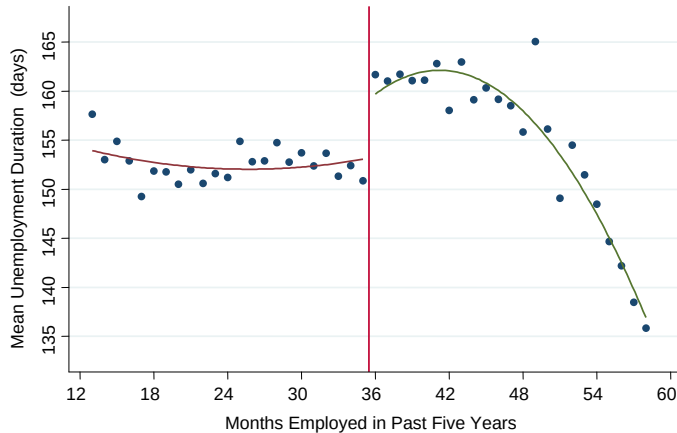


Empirical Estimation of Effects of UI: Evidence

- Meyer (1990) and many others implement this method using data on unemployment durations in the U.S. and state-level reforms
- General finding: benefit elasticity of 0.4-0.6
- 10% rise in unemployment benefits leads to about a 4-6% increase in unemployment durations.
- More recent empirical approach: regression discontinuity
- Card-Chetty-Weber (2007) use the fact that in Austria, you get up to 30 weeks of benefits when you have been employed for 36+ months in last 5 years (instead of up to 20 weeks)
- Can look at duration of unemployment based on how long you have worked in last 5 years
⇒ Finds somewhat smaller elasticity around 0.3

Card, Chetty, Weber (2007)

Effect of Benefit Extension on Unemployment Durations

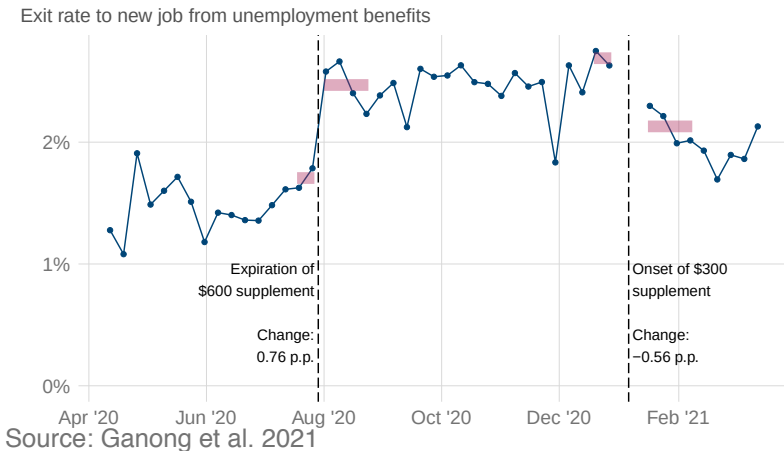


Effects of UI Expansion during COVID

- Ganong et al. (2021) analyze the impact of the huge UI expansion during COVID using JP Morgan bank data
- They find that weekly exit rate from UI to new job:
 - Jumps up from 1.8% to 2.6% when \$600 supplement ends
 - Jumps down from 2.6% to 2.0% when \$300 supplement starts
- $\Rightarrow$ Clear negative moral hazard effects of UI
- But quantitative effect appears relatively small (difficult to see in aggregate unemployment rate during recovery). Likely efficient way to help job losers during COVID.

Figure 2: Effect of Expanded Benefits on Job-Finding: Interrupted Timeseries Design

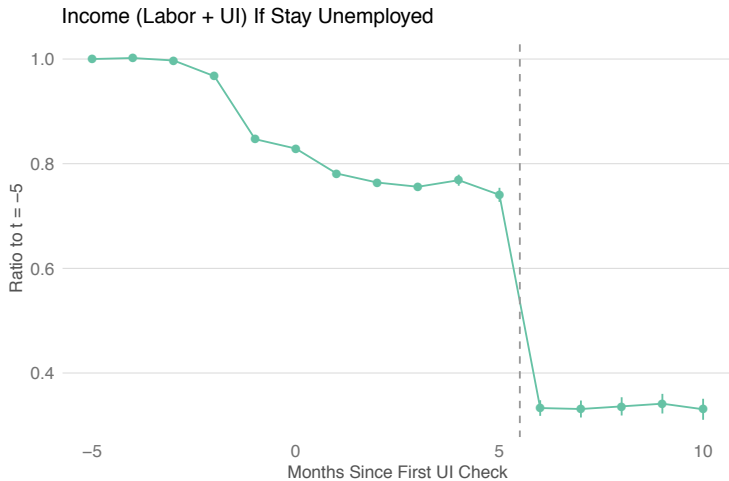
(a) Interrupted Timeseries Estimate

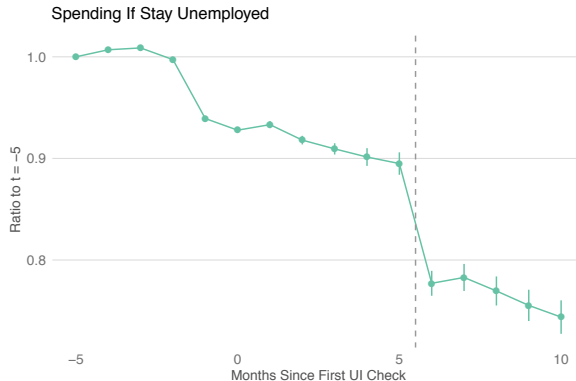


Evidence on Consumption-Smoothing

- Difference-in-difference strategy has been used to examine how UI benefits affects consumption
- Gruber (1997) finds that consumption falls on average when people lose their job by about 10-15%
- \$1 increase in UI benefits increases consumption by 30 cents
- Much less than 1-1 because savings behavior changes, spousal labor supply, borrowing from friends, etc. (this is called self-insurance)
- Recent study by Ganong-Noel AER'19 uses bank account data to follow people through UI spell $\Rightarrow$ Finds big effects of UI benefit exhaustion on consumption especially for groups with high replacement rates or low wealth

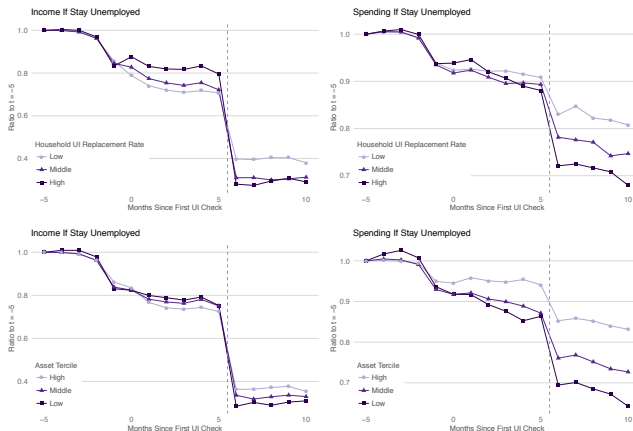
Figure 2: Income and Spending If Stay Unemployed





Notes: This figure plots income and spending for the sample that stays unemployed. In months $t = \{-5, -4, -3, -2, -1, 0\}$, this includes everyone who receives UI at date 0 and meets the sampling criteria described in Section 2.1. In month $t = 1$, this includes only households who continue to receive UI and excludes households who receive their last UI check in month 0. In month $t = 2$, this excludes households who receive their last UI check in month 0 or month 1, and so on. Employment status after UI exhaustion is measured using paycheck deposits. The vertical line marks UI benefit exhaustion. Income is positive after UI benefit exhaustion because of labor income of other household members. Vertical lines denote 95 percent confidence intervals for change from the prior month. See Section 3.1.1 for details.

Figure 3: Heterogeneity in Income and Spending If Stay Unemployed



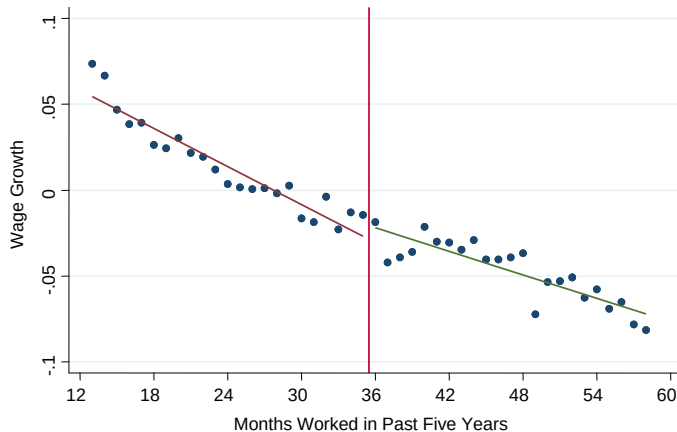
Notes: This figure shows heterogeneity in income and spending by the ratio of UI benefits to estimated household annual income and the ratio of estimated total liquid assets (a measure described in Section 2.2) to consumption prior to the onset of unemployment. The sample is households that receive UI and stay unemployed, as described in the note to Figure 2.

Does UI Have Long-Term Benefits?

- Another potential benefit of UI, neglected in simple model above: improvements in match quality
- Are people forced to take worse jobs because they have to rush back to work to put food on the table? E.g. engineer starts working at McDonalds.
- Can examine this using similar data
- Look at whether people who got higher benefits and took longer to find a job are better off years later
- Card-Chetty-Weber (2007) exploit again the regression discontinuity and find no long-term match benefit on subsequent wage or subsequent job duration

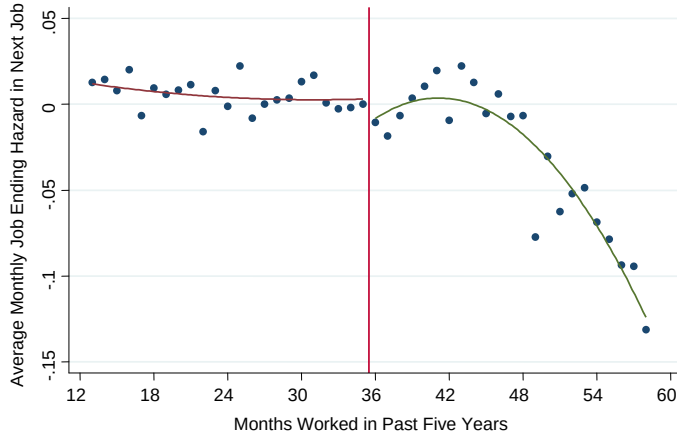
Card, Chetty, Weber (2007)

Effect of Extended Benefits on Subsequent Wages



Card, Chetty, Weber (2007)

Effect of Extended Benefits on Subsequent Job Duration



Summary of Empirical Findings on UI

- Higher benefit level $\Rightarrow$ longer unemployment durations (moral hazard cost)
- Higher benefit level $\Rightarrow$ more consumption while unemployed (consumption smoothing benefit)
- UI benefits have no beneficial effects on long-term job outcomes
- $\Rightarrow$ Model implies that providing some UI is desirable but not full replacement of lost wages. UI replacement rate should be around 50% based on those empirical findings.

Should UI Benefits Be Extended during Recessions?

- US extends UI benefits during recessions. Extensions ended in 2014 (controversial policy debate)
- Social Justice: Harder to find jobs in recessions $\Rightarrow$ being unemployed is less of a choice $\Rightarrow$ Extending benefits is desirable
- Efficiency: In recessions, the job market is too slack [too hard to find jobs, too easy for firms to find workers].
 - If longer UI benefits decrease slack in labor market then longer UI benefits desirable [this is the case if UI benefits stimulate aggregate demand or if job seekers compete for a fixed number of jobs in recession]
 - If longer UI benefits increase slack in labor market then shorter UI benefits desirable [this is the case if longer UI benefits increase the bargaining power of workers and hence increase wages further reducing labor demand]
- Economists try to tell apart these two cases using empirical evidence

Disability Insurance

- Disability is conceptually close to retirement: some people become unable to work before old age (due to accidents, medical conditions, etc.)
- All advanced countries offer public disability insurance almost always linked to the public retirement system
- Disability insurance allows people to get Social Security retirement benefits before the “Early Retirement Age” if they are unable to work due to disability

US Disability Insurance

- Federal program funded by OASDI payroll tax, pays SS benefits to disabled workers under retirement age (similar computation of benefits based on past earnings)
- Program started in 1956 and became more generous overtime (age 50+ condition removed, definition of disability liberalized, replacement rate has grown)
- Eligibility: Medical proof of being unable to work for at least a year, need some prior work experience, 5 months waiting period with no earnings required (screening device)
- Social security examiners rule on applications. Appeal possible for rejected applicants. Imperfect process with big type I and II errors (Parsons AER'91) $\Rightarrow$ Scope for Moral Hazard
- DI tends to be an absorbing state (most beneficiaries won't ever work again). Can earn up of \$1200/month while on DI.

- In 2021, about 9.2m DI beneficiaries (not counting widows+children), about 5-6% of working age (20-64) population
- Very rapid growth: In 1960, less than 1% of working age population was on DI
- Growth particularly strong during recessions: early 90s, late 00s. Slight decline from 10.3m in 2013 to 9.2m in 2021
- Key empirical question: Are DI beneficiaries unable to work? or are DI beneficiaries not working because of DI.

Chart 2.

All Social Security disabled beneficiaries in current-payment status, December 1970–2017

The number of disabled beneficiaries has risen from 1,812,786 in 1970 to 10,059,166 in 2017, driven predominately by an increase in the number of disabled workers. The number of disabled adult children has grown slightly, and the number of disabled widow(er)s has remained fairly level. In December 2017, there were 8,695,475 disabled workers; 1,105,405 disabled adult children; and 258,286 disabled widow(er)s receiving disability benefits.

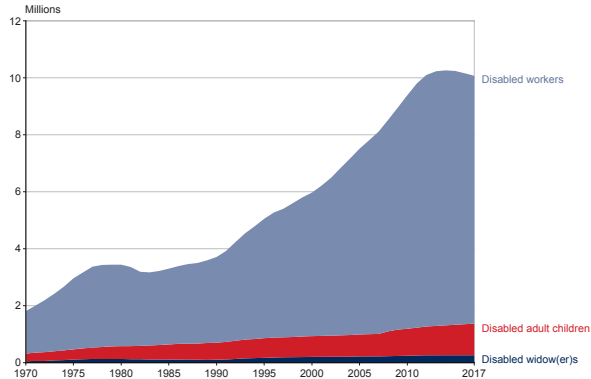
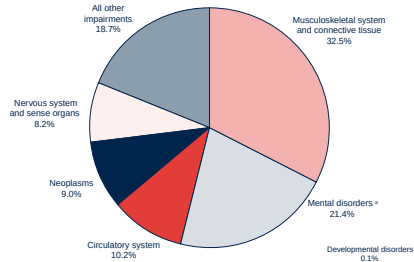


Chart 10.
Disabled-worker awards, by selected diagnostic group, 2010

Source: SSA DI annual report

In 2010, 1,026,988 disabled workers were awarded benefits. Among those awardees, the most common impairment was diseases of the musculoskeletal system and connective tissue (32.5 percent), followed by mental disorders (21.4 percent), circulatory problems (10.2 percent), neoplasms (9.0 percent), and diseases of the nervous system and sense organs (8.2 percent). The remaining 18.7 percent of awardees had other impairments.



- Detecting disability is challenging, particularly for back injuries and mental health conditions
- One way to quantify difficulty in assessment: audit study
- Take a set of disability claims that was initially reviewed by a state panel
- One year later, resubmit them to the panel as anonymous new claims.
- Compare decisions on the same cases
- $\Rightarrow$ Substantial evidence of Type I errors (incorrect rejection of a disabled person) and Type II errors (letting a non-disabled person on the program)

TABLE 1—REASSESSMENTS OF INITIAL
SOCIAL SECURITY DETERMINATIONS

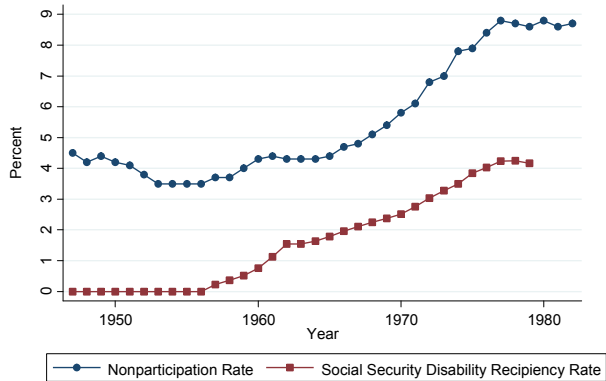
*A. Bureau of Disability Insurance Review One Year
After Initial Determination (Percentages):*

BDI assessment	Initial determination	
	Allowance	Denial
Allowance	78.8	21.1
Denial	22.5	77.5

Note: The sample sizes are 250 initial allowances and 248 initial denials.

Source: Smith and Lilienfeld (1971 p. 195).

Nonparticipation and Reciprocity Rates, Men 45-54 Years Old



Source: Parsons 1984 Table A1

DI Empirical Effects: Observational Studies

- Parallel growth of DI recipients and non-participation rates among men aged 45-54 but causality link not clear
- Cross-Sectional Evidence (Parsons '80): Does potential DI replacement rate have an impact on labor force participation (LFP) decision?
- Uses cross-sectional variation in potential replacement rates
- Survey data on men aged 45-59 from 1966-69
- OLS regression: $NLFP_i = \alpha + \beta \cdot DIreplate_i + \varepsilon_i$
- Large effect that can fully explain decline in LFP among men 45+

- Issues with Cross-Sectional Evidence:
 - ① $Dlreprate_i$ depends on wages (higher for low wage earners) and likely to be correlated with ε_i (likelihood to become truly disabled)
 - ② Impossible to control fully for wages in regression because all variation in $Dlreprate_i$ is due to wages
 - ③ Bound AER'89 replicates Parson's regression on sample that never applied to DI and obtains similar effects implying that the OLS correlation not driven by UI

DI Empirical Effects: Rejected Applicants

- Bound AER'89 proposes a technique to bound effect of DI on LFP rate
- Uses data on LFP on (small sample of) rejected applicants as a counterfactual
- Idea: If rejected applicants do not work, then surely DI recipients would not have worked
⇒ Rejected applicants' LFP rate is an upper bound for LFP rate of DI recipients absent DI
- Results: Only 30% of rejected applicants return to work and they earn less than half of the mean non-DI wage
- ⇒ at most $1/3$ of the trend in male LFP decline can be explained by shift to DI
- Von Waechter-Manchester-Song AER'11 replicate Bound using full pop SSA admin data and confirm his results

TABLE 2—EMPLOYMENT, EARNINGS, AND OTHER CHARACTERISTICS OF REJECTED DISABILITY INSURANCE APPLICANTS

	1972			1978		
	Population	Rejected Applicants	Beneficiaries	Population	Rejected Applicants	Beneficiaries
Labor Supply						
Percent Employed	77.7	32.6	3.2	69.3	28.7	2.3
Percent Worked 71/77	91.9	45.0	7.5	86.7	40.4	5.5
Percent Full Year						
(≥ 50 Weeks) ^a	76.8	47.4	31.4	83.5	41.2	22.2
Percent Full Time						
(≥ 35 Hours) ^a	95.4	75.9	25.0	92.4	79.6	38.3
Earnings Among Positive Earners						
Median Annual Earnings, 71/77 ^b	\$9000	\$4000	\$700	\$14000	\$5300	\$1000

Source: Bound 1991

DI Empirical Effects: Rejected Applicants

- Maestas-Mullen-Strand AER'13 obtain causal effect of DI on LFP using natural variation in DI examiners' stringency and large SSA admin data linking DI applicants and examiners
- Idea: (a) Random assignment of DI applicants to examiners and (b) examiners vary in the fraction of cases they reject $\Rightarrow$ Valid instrument of DI receipt
- Result 1: DI benefits reduce LFP of applicants by 28 points $\Rightarrow$ DI has moderate impact (consistent with Bound AER'89)
- Result 2: DI has heterogeneous impact: small effect on those severely impaired but big effect on less severely impaired
- Tough judges marginal cases unlikely to work without DI, lenient judges marginal cases somewhat likely to work without DI

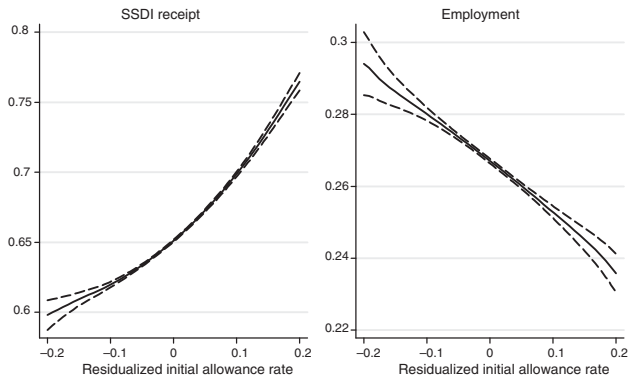


FIGURE 4. SSDI RECEIPT AND LABOR SUPPLY BY INITIAL ALLOWANCE RATE

Notes: Ninety-five percent confidence intervals shown with dashed lines. Employment measured in the second year after the initial decision. Bandwidth is 0.116 for DI and 0.130 for labor force participation.

Source: DIODS data for 2005 and 2006

DI Generosity Effects: Regression Kink Design (RKD)

- DI benefits calculated like SS benefits: AIME formula based on average life-time earnings creates a “kinked” relationship
- Ideal setting for an RKD (Card et al. 2015): test whether outcome such as earnings or mortality is also “kinky”
 - ① Test first for no sorting of DI recipients around kink to validate RKD design [similar to RDD validation]
 - ② RKD estimate: $\text{Change in slope of outcome at kink} / \text{Change in slope of benefits at kink}$
- Gelber et al. '17 analyze effects on earnings of DI generosity and find an income effect of -\$0.2 per dollar of benefits
- Gelber et al. '18 analyze effects on mortality: at lower bend point, \$1K extra DI/year reduces annual mortality by .25 points (1 out of 400 lives saved)

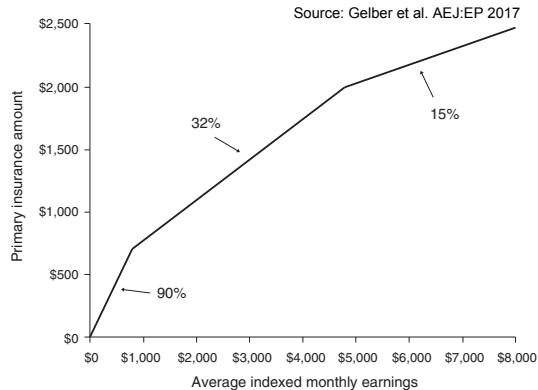


FIGURE 1. PRIMARY INSURANCE AMOUNT AS A FUNCTION OF AVERAGE INDEXED MONTHLY EARNINGS

Notes: The figure shows the primary insurance amount (PIA) as a function of average indexed monthly earnings (AIME) in 2013. The percentages are marginal replacement rates.

Source: SSA (2013)

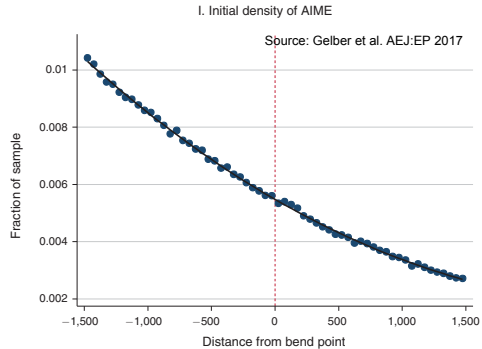


FIGURE 3. SMOOTHNESS OF DENSITY AND PREDETERMINED COVARIATES AROUND THE UPPER BEND POINT (*continued*)

Notes: The figure shows the density of initial AIME in \$50 bins as a function of distance of initial AIME to the upper bend point. The number of observations appears smooth through this bend point, with no sharp change in slope or level. The upper bend point is where the marginal replacement rate in converting AIME to PIA changes from 32 percent to 15 percent. The sample includes DI beneficiaries within \$1,500 of the upper bend point (see the text for other sample restrictions). The fraction of the sample in each bin is calculated by dividing the number of beneficiaries in each bin by the total number of beneficiaries in the sample. The best-fit line is a ninth-order polynomial that parallels the regression presented in Table 2 that minimizes the corrected Akaike Information Criterion (AICc).

Source: The data are from SSA administrative records.

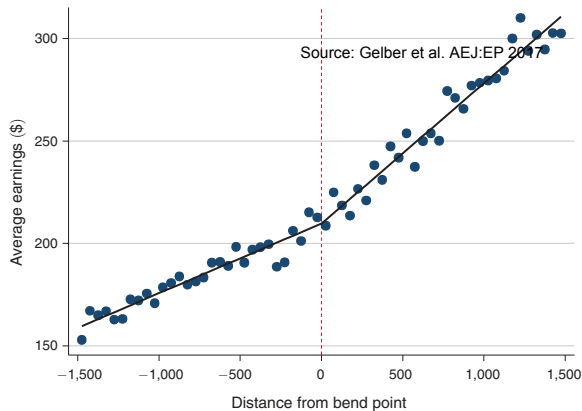
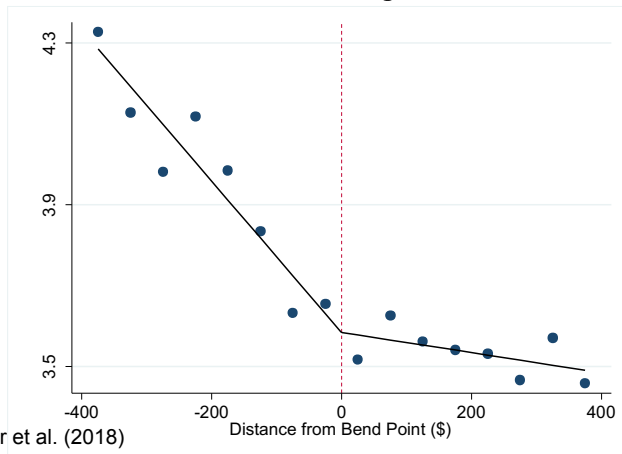


FIGURE 4. AVERAGE MONTHLY EARNINGS AFTER DI ALLOWANCE

Notes: The figure shows mean monthly earnings in the first four years after going on DI, in \$50 bins, as a function of distance of AIME from the bend point, where AIME is measured when applying for DI. The figure shows that mean earnings slope upward more steeply above the upper bend point than below it, with fitted lines that lie close to the data.

Figure 3. Annual Percent Mortality Rates around the Bend Points
A: Lower bend point



Source: Gelber et al. (2018)

Workers' Compensation: Institutional Features

- Workers' compensation is insurance for injuries on the job, mainly temporary injuries that prevent work (short-term)
- Workers' Compensation is a state-level program
- Two components: medical and indemnity
- Indemnity payment replaces roughly two-thirds of lost wages.
- Unlike UI, WC payments are untaxed, leading to a higher replacement that is near 90% on average.
- Substantial variation across states in benefit levels

Workers' Compensation (WC): Institutional Features

- Workers' comp is a mandated benefit; no explicit tax but firms required by law to provide this benefit to workers
 - Most firms choose to buy coverage from private insurers
 - Premiums are more tightly experience rated than UI because they are determined by private sector
 - Insurance companies charge high-risk firms more.
- Important feature of WC: no-fault insurance.
 - When there is a qualifying injury, WC benefits paid regardless of whether the injury was the worker's or the firm's fault.
 - Idea: reduce inefficiency of tort system (legal costs) by having fixed rules and not worrying about liability

Moral Hazard in Workers' Compensation

- Moral hazard in WC can manifest itself in reported injuries, injury durations, and types of injuries reported.
- E.g. easier to report back pain—very hard to verify
- Huge issue in CA—companies paid high workers comp rates
- Governor Schwarzenegger reform in 2004 cut benefits sharply, claiming to reduce injuries and “open CA for business”
- Is it true that there is substantial moral hazard? Again, consider several pieces of evidence
- Strategy 1: Timing of injuries. “Monday effect” (faking week-end injuries into work injuries)

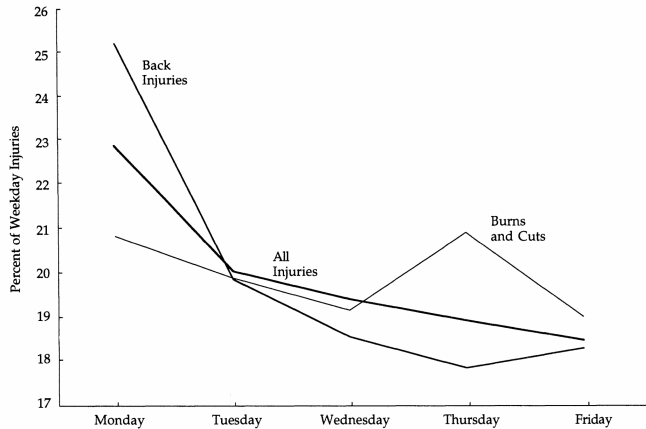


Figure 1. Distribution of Weekday Injuries.

Source: Card and McCall 1996

Moral Hazard in Workers' Compensation

- Strategy 2: examine effect of workers comp benefit levels on durations using a diff-in-diff strategy (Meyer, Viscusi, Durbin 1995)
- Reforms in Kentucky and Michigan that increased benefits for high-earning workers (but not low-earning workers) in late 1980s
- Compare changes in injury durations and medical costs for high-earners vs. low earners in those states before and after reform

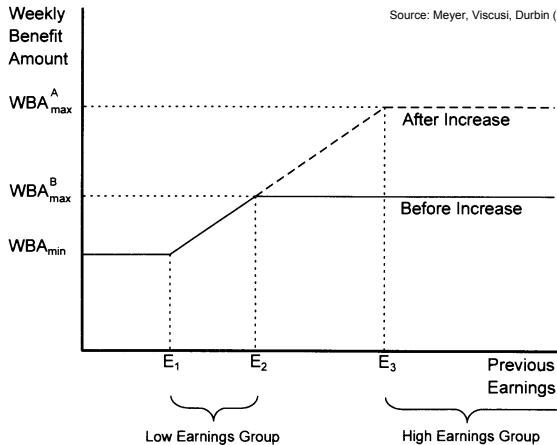


FIGURE 1. TEMPORARY TOTAL BENEFIT SCHEDULE
BEFORE AND AFTER AN INCREASE IN
THE MAXIMUM WEEKLY BENEFIT

Variable	Kentucky			Michigan		
	Before increase (1)	After increase (2)	Percentage change (3)	Before increase (4)	After increase (5)	Percentage change (6)
Maximum benefit (\$)	131.00	217.00	65.65	181.00	307.00	69.61
Replacement rate, high earnings (percent)	32.70 (0.25)	51.02 (0.37)	56.02 (1.65)	30.01 (0.35)	44.15 (0.48)	47.14 (2.33)
Replacement rate, low earnings (percent)	66.42 (0.20)	66.66 (0.22)	0.36 (0.44)	66.64 (0.24)	66.35 (0.30)	-0.45 (0.58)

Source: Meyer, Viscusi, Durbin 1995

TABLE 4—KENTUCKY AND MICHIGAN: DURATION AND MEDICAL COSTS OF TEMPORARY TOTAL DISABILITIES
DURING THE YEARS BEFORE AND AFTER BENEFIT INCREASES

Variable	High earnings		Low earnings		Differences		Difference in differences
	Before increase (1)	After increase (2)	Before increase (3)	After increase (4)	[(2) – (1)] (5)	[(4) – (3)] (6)	[(5) – (6)] (7)
Mean duration (weeks)							
Kentucky	11.16 (0.83)	12.89 (0.83)	6.25 (0.30)	7.01 (0.41)	1.72 (1.17)	0.76 (0.51)	0.96 (1.28)
Michigan	14.76 (2.25)	19.42 (2.67)	10.94 (1.09)	13.64 (1.56)	4.66 (3.49)	2.70 (1.90)	1.96 (3.97)
Median duration (weeks)							
Kentucky	4.00 (0.14)	5.00 (0.20)	3.00 (0.11)	3.00 (0.12)	1.00 (0.25)	0.00 (0.16)	1.00 (0.29)
Michigan	5.00 (0.45)	7.00 (0.67)	4.00 (0.22)	4.00 (0.28)	2.00 (0.81)	0.00 (0.35)	2.00 (0.89)
Median medical cost (dollars)							
Kentucky	393.51 (19.29)	411.49 (22.72)	238.96 (8.48)	254.40 (9.11)	17.98 (29.80)	15.44 (12.44)	2.55 (32.30)
Michigan	689.73 (77.30)	765.00 (134.53)	390.63 (32.80)	435.00 (33.09)	75.27 (155.16)	44.38 (46.59)	30.89 (162.00)

Source: Meyer, Viscusi, Durbin 1995

Moral Hazard in Workers' Compensation

- Result: 10% increase in WC benefit raises out-of-work duration due to injury by 4%
- Again, need to weigh this against benefits to reach policy conclusions
- Give people more time to heal after injury without rushing them back to work
- Higher consumption while out of work
- No evidence yet on these issues

Poll (No Right Answer)

A person who is poor because they just lost their job should get a larger transfer than someone who is equally poor and has been that poor for a long time.

- A. True
- B. False
- C. Uncertain

When the government gives transfers to unemployed people, they are less likely to find a job.

- A. True
- B. False
- C. Uncertain

- Individuals clearly value the consumption smoothing provided by social insurance programs
- In each case there are moral hazard costs associated with the provision of the insurance
- Empirical analyses of all three programs can be used to inform policy makers' decisions as program reforms move forward

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